

BYUNG HYUN LEE

CONTACT INFORMATION

Affiliation: Department of Electrical and Computer Engineering, Seoul National University

 Email  Website  Google Scholar  LinkedIn

RESEARCH INTERESTS

My research focuses on *selective knowledge updates* for deep visual and foundation models — adding new knowledge or removing unwanted knowledge while preserving existing capabilities, an essential challenge for efficient and trustworthy AI. I have studied this problem across online continual learning for agentic and physical AI, continual learning for histopathology as a vertical AI domain, and concept erasure for visual generative models, through methods such as hierarchy- and future-aware learning, orthogonal low-rank adaptation, and gated, spatially localized updates. Recently, I am extending these ideas to multimodal LLMs, agentic AI, and vision-language-action models, toward efficient and reliable knowledge updates for omnimodal and vertical foundation models in real-world settings.

Additionally, I'm interested in model acceleration, image restoration, and histopathological analysis.

EDUCATION

- **Seoul National University (SNU)** *Mar 2021 - Feb 2027 (expected)*
Combined M.S./Ph.D Program in Electrical and Computer Engineering (ECE)
- **Ulsan National Institute of Science and Technology (UNIST)** *Mar 2015 - Aug 2018*
B.S. in Electrical Engineering, GPA: 4.04/4.30

RESEARCH ADVISOR

- **Se Young Chun:** Professor, Department of Electrical and Computer Engineering (ECE), SNU

PUBLICATIONS (Top AI Conference Papers)

- [Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models](#)
H. Seo*, **B. H. Lee***, J. Cho, S. Lim, S. Y. Chun
Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis](#)
B. H. Lee, W. Jeong, W. Han, K. Lee, S. Y. Chun
International Conference on Computer Vision (ICCV), 2025
- [Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation](#)
B. H. Lee*, S. Lim*, S. Y. Chun (*co-first authors)
Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- [Concept pinpoint eraser for text-to-image diffusion models via residual attention gate](#)
B. H. Lee*, S. Lim*, S. Lee, D. U. Kang, S. Y. Chun (*co-first authors)
International Conference on Learning Representations (ICLR), 2025
- [Doubly perturbed task free continual learning](#)
B. H. Lee, M. Oh, S. Y. Chun
Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, oral), 2024

- [Online Continual Learning on Hierarchical Label Expansion](#)
B. H. Lee*, O. Jung*, J. Choi, S. Y. Chun (*co-first authors)
International Conference on Computer Vision (ICCV), 2023
- [All-in-one image restoration for unknown degradations using adaptive discriminative filters for specific degradations](#)
D. Park, B. H. Lee, S. Y. Chun
Conference on Computer Vision and Pattern Recognition (CVPR), 2023

PUBLICATIONS (Journal Papers)

- [Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation](#)
H. Lee, H. Lee, B. H. Lee, S. Y. Chun
IEEE Access, 2025
- [Towards accelerating model parallelism in distributed deep learning systems](#)
H. Choi*, B. H. Lee*, S. Y. Chun, J. Lee (*co-first authors)
PLOS One, 2023

PUBLICATIONS (Short, Workshop, or Domestic Papers)

- [Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models](#)
K. R. Lee*, K. H. Lee*, S. Hong, B. H. Lee, S. Y. Chun
CVPR Workshop on Machine Unlearning for Computer Vision, 2026
- [Selective Concept Erasing for Safe Diffusion Models](#)
B. H. Lee, S. Lim, S. Y. Chun
Korea Signal Processing Conference (Best Poster Presentation Award), 2024
- [Expert classifier ensemble based post-processing correction for unbiased scene graph generation](#)
S. Lee, B. H. Lee, S. Y. Chun
Workshop on Image Processing and Image Understanding (IPIU), 2024
- [Efficient and accurate quantized image super-resolution on mobile npus, mobile ai & aim 2022 challenge: report](#)
A. Ignatov et al. (including B. H. Lee)
European Conference on Computer Vision Workshops (ECCVW), 2022
- [Efficient single-image depth estimation on mobile devices, mobile AI & AIM 2022 challenge: report](#)
A. Ignatov et al. (including B. H. Lee)
European Conference on Computer Vision Workshops (ECCVW), 2022
- [Uncertainty-based dual domain low-dose X-ray CT reconstruction](#)
S. Lee, D. U. Kang, B. H. Lee, S. Y. Chun
Korean Signal Processing Conference, 2022
- [Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems in Image Processing](#)
B. H. Lee, S. Y. Chun
International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021

TECHNICAL REPORTS / MANUSCRIPTS UNDER REVIEW

- [Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs](#)
Under review, 2026
- [Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis](#)
Under review, 2026
- [HYPERION: Joint-Hierarchy Hyperbolic Fine-Tuning for Histopathology Slide Representations](#)
Under review, 2026

- [RePath: Zero-Shot Tumor Segmentation in Gigapixel WSIs via Differential Intra-Slide Patch Affinity](#)
Under review, 2026
- [HyperCLOVA X 8B Omni](#)
HyperCLOVA X Team in NAVER Cloud (including **B. H. Lee**)
2026
- [HyperCLOVA X 32B Think](#)
HyperCLOVA X Team in NAVER Cloud (including **B. H. Lee**)
2026
- [Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation](#)
H. Seo*, J. Bang*, H. Lee*, J. Lee, **B. H. Lee**, S. Y. Chun
arXiv, 2025

PATENTS

- **Training-free Industrial Defect Segmentation.**
S. Y. Chun, D. U. Kang, H. Kim, J. H. Jang, **B. H. Lee** & Hoigi Seo
Korean Patent, Filed. 2025.
- **All-in-one image quality improvement model providing method performing image quality restoration for multiple image quality degradation factors**
S. Y. Chun, D. Park, **B. H. Lee**
U.S. Patent, Filed, 2023
- **Method for providing all-in-one image quality improvement model that performs image quality restoration for multiple image quality inhibitors**
S. Y. Chun, D. Park, **B. H. Lee**
Korea Patent, Filed, 2023

EXPERIENCES

- **National AI Foundation Model Project in South Korea**, NAVER Cloud *Oct 2025 - Feb 2026*
Visiting Student Researcher
- **Biomedical Medical Image Processing Lab**, UNIST *May 2020 - Feb 2021*
Researcher (Advisor: Prof. Se Young Chun)
- **Biomedical Medical Image Processing Lab**, UNIST *July 2017 - Aug 2018*
Student Internship (Advisor: Prof. Se Young Chun)

PRESENTATIONS

- **Continual Learning and Its Applications in Magnetic Resonance Imaging**
Advanced neuroimaging and AI workshop, SNU, 2024

AWARDS AND HONORS

- **Apple Scholars in AI/ML PhD Fellowship Program** *2026*
Institutional Nominee
- **Qualcomm Innovations Fellowship Korea** *2025*
Best paper finalist
- **2nd Place in Report Generation in Pathology using Pan-Asia Gigapixel WSIs** *2025*
Challenge by International Conference on MICCAI (Served as the team leader, 26 teams participated)

- **Yulchon AI Star Scholarships** (₩8,000,000)
Youlchon Foundation & SNU AI Institute

2025

PROJECTS

- **AI for Human-Machine Teaming**
United States Air Force & IITP, Republic of Korea *Mar 2025 - Present*
- **Industrial Defect Segmentation**
Samsung Device Solution (DS) *Mar 2024 - Mar 2025*
- **AI System for Gigapixel Histopathological WSI Analysis**
Ministry of Health and Welfare, Republic of Korea *Aug 2023 - Dec 2024*
- **Relations and Affordance Aware Robot Grasping**
Samsung Research, Republic of Korea *May 2022 - May 2023*

EXTRACURRICULAR EXPERIENCES

- **Military Service, Republic of Korea Army**
Discharged as Sergeant *Sep 2018 - Apr 2020*
- **Pinocchio (Robot Club), UNIST**
Education Director *Mar 2015 - Aug 2018*